Yuyou Zhang

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RESEARCH INTERESTS

- **Vision–Language Models (VLMs):** Spatial and temporal reasoning, perspective taking, 3D world understanding, and diagnostic benchmark design for multimodal models.
- **Large Language Models (LLMs):** Safety, alignment, and responsible AI, including reasoning-based safety mechanisms.
- **Embodied AI & Reinforcement Learning:** Learning and control for embodied agents, with a focus on robust decision-making, adaptability, and safety in dynamic real-world environments.

EDUCATION

Carnegie Mellon University — **Pittsburgh, USA**
Ph.D. in Mechanical Engineering, Robotics focus, Safe AI Lab — *Aug. 2022 - May 2027 (expected)*

Shanghai Jiao Tong University — **Shanghai, China**
B.Eng., Automation, School of Electronic Information Electrical Engineering
B.Eng., ZhiYuan Honors Program, Zhiyuan College — *Sept. 2018 - June 2022*

SELECTED PUBLICATIONS

[1] **Y. Zhang**, R. Corcodel, C. Hori, A. Cherian, and D. Zhao, "Spinbench: Perspective and rotation as a lens on spatial reasoning in vlms," *Best Paper Runner Up NeurIPS SPACE in Vision, Language, and Embodied AI Workshop*, 2025.

[2] **Y. Zhang**, M. Li, W. Han, Y. Yao, Z. Cen, and D. Zhao, "Safety is not only about refusal: Reasoning-enhanced fine-tuning for interpretable llm safety," *The 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.

[3] **Y. Zhang**, Y. Yao, S. Liu, Y. Niu, C. Lin, Y. Yang, W. Yu, T. Zhang, J. Tan, and D. Zhao, "Quietpaw: Learning quadrupedal locomotion with versatile noise preference alignment," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.

[4] **Y. Zhang**, R. Corcodel, and D. Zhao, "Bipedalism for Quadrupedal Robots: Versatile Loco-Manipulation through Risk-Adaptive Reinforcement Learning," in *IEEE-RAS International Conference on Humanoid Robots*, Sept. 2025.

[5] **Y. Zhang**, Y. Niu, X. Liu, and D. Zhao, "Composer: Scalable and robust modular policies for snake robots," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 10800–10806, IEEE, 2024.

[6] C. Wang, **Y. Zhang**, X. Zhang, Z. Wu, X. Zhu, S. Jin, T. Tang, and M. Tomizuka, "Offline-online learning of deformation model for cable manipulation with graph neural networks," *IEEE Robotics and Automation Letters (RAL)*, vol. 7, no. 2, pp. 5544–5551, 2022.

[7] L. Han, **Y. Zhang**, and H. Wang, "Vision-based contact point selection for the fully non-fixed contact manipulation of deformable objects," *IEEE Robotics and Automation Letters (RAL)*, vol. 7, no. 2, pp. 4368–4375, 2022.

[8] L. Han, **Y. Zhang**, and H. Wang, "Hybrid adaptive vision-force control under the bottleneck constraint," *IEEE Transactions on Control Systems Technology*, 2022.

[9] F. Xu, **Y. Zhang**, J. Sun, and H. Wang, "Adaptive visual servoing shape control of a soft robot manipulator using bézier curve features," *IEEE/ASME Transactions on Mechatronics*, 2022.

RESEARCH EXPERIENCE

Safe AI Lab — **Carnegie Mellon University**
Ph.D Candidate — *Aug 2022 - present*

SpinBench: Perspective and Rotation as a Lens on Spatial Reasoning in VLMs [1]

- SpinBench is a cognitively grounded diagnostic benchmark for evaluating spatial reasoning in vision language models (VLMs).
- SpinBench targets translation, rotation, object relative pose, and viewpoint change.
- Progressively structured: single-object tasks scaffold toward multi-object perspective-taking tasks.

Safety is Not Only About Refusal: Reasoning-Enhanced Fine-tuning for Interpretable LLM Safety [2]

- Proposed a novel framework that trains models to engage in explicit safe reasoning before response.
- Reasoning is not only a key component of model capability but also powerful for safety alignment.

QuietPaw: Learning Quadrupedal Locomotion with Versatile Noise Preference Alignment [3]

- Proposed a conditional RL framework for noise-constrained locomotion.
- A decomposed value estimation approach that improves Pareto-optimality.

Bipedalism for Quadrupedal Robots: versatile loco-manipulation through risk-adaptive RL [4]

- Proposed a risk-adaptive RL framework for the robust bipedal locomotion of quadrupedal robots
- Introduced an uncertainty metric based on the return distribution for risk awareness
- Robust loco-manipulations with cart pushing, contact-aware obstacle probing, payload carrying

COMPOSER: scalable and robust modular policies for snake robots [5]

- Proposed a modular policy with a self-attention mechanism to enhance cooperative behaviors
- Demonstrated superior efficiency, robustness against corruption, and zero-shot generalizability.

Mechanical Systems Control Lab — **University of California, Berkeley**

Undergraduate Researcher — *June 2021 - Nov. 2021*

Offline-online learning of deformation model for cable manipulation with graph neural networks [6]

- Combined offline graph neural network with online residual model to approximate cable dynamics
- Improved model training efficiency and generalizability, narrowed sim-to-real gap.

Intelligent Robotics and Machine Vision Lab — **Shanghai Jiao Tong University**

Undergraduate Researcher — *Sept. 2020 - June 2022*

Safe Vision-based contact selection for the non-fixed contact manipulation of deformable objects[7]

- Proposed a stabilizing and optimization strategy to select initial manipulation contact points
- Designed an uncalibrated visual servo controller to validate the contact optimization strategy

Hybrid vision-force control for robotics manipulation in confined space[8]

- Developed an adaptive method to estimate Jacobian matrixes in force space and image space
- Designed a hybrid vision-force controller for manipulation with a desired contact force

Adaptive visual servoing shape control of a soft robot manipulator using *Bézier* curve features[9]

- Proposed an adaptive *Bézier* curve update algorithm to represent soft continuum robots
- Designed an uncalibrated visual servo controller for shape regulation of the soft continuum robot

WORKING EXPERIENCE

Mitsubishi Electric Research Laboratories

Research Intern — *May. 2025 - Aug. 2025*

- Proposed **SpinBench**, a diagnostic benchmark for spatial reasoning in VLMs.
- Conducted large-scale evaluation of **43+ VLMs** across spatial transformation tasks.
- **Fine-tuned InternVL** on curated data, improving spatial reasoning performance.

Mitsubishi Electric Research Laboratories

Research Intern — *May. 2024 - Aug. 2024*

- Developed distributional risk-aware policy for quadrupedal robot locomotion
- Proposed bipedal locomotion for quadrupedal robots to enable flexible loco-manipulation.

Flexiv Robotics Inc.

Robotic Vision Intern — *Dec. 2021 - Mar. 2022*

- 3D reconstruction and point cloud registration
- Point cloud feature extraction and visual fixture generation for interactive teleoperation

Shanghai Inspection and Testing Institute of Instruments and Automation Systems Co. Ltd.

Engineer Intern — *June. 2021 - Aug. 2021*

TECHNICAL SKILLS

ML: Vision–Language Models, Large Language Models, Multimodal Learning, Model Evaluation, Finetuning, Benchmark Design, Reinforcement Learning, Distributional RL, Imitation Learning
Frameworks: PyTorch, HuggingFace
Robotics: Locomotion, Loco-manipulation, Visual Servoing

AWARDS

- CMU Graduate Student Assembly Travel Grants for ICRA 2024
- Carnegie Institute of Technology Dean's Fellowship, 2022
- Shanghai Outstanding Graduate, Top 2.5%
- Zhiyuan Honors Degree, Top 1.5%
- Zhiyuan Honors Scholarship, 2018, 2019, 2020, 2021, Top 5%
- Undergraduate Excellence Scholarship 2018, 2019, 2020, 2021
- The Mathematical Contest in Modeling: Meritorious Winner, Top 3%